Year 9 · Chapter 10 · Surface area and volume: answers (teacher copy)

10.1 Metric units (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: cm → mm: 14 × 10 = 140 mm
2. Ex 1b: cm → m: 250 ÷ 100 = 2.5 m
3. Ex 1c: kg → g: 3.5 × 1000 = 3500 g
4. Ex 1d: mL → L: 750 ÷ 1000 = 0.75 L
5. Ex 2a: h → min: 2.5 × 60 = 150 min
6. Ex 2b: GB → MB: 4.2 × 1000 = 4200 MB
7. Ex 2c: km → m → cm: × 1000 then × 100, so 1.5 × 1000 × 100 = 150 000 cm
8. Ex 2d: mm → µm: 0.5 × 1000 = 500 µm

Easy

1. 90 mm
2. 200 cm
3. 4000 m
4. 600 cm
5. 12 cm
6. 3 m
7. 5 km
8. 2 m
9. 2000 g
10. 4 kg
11. 1500 g
12. 2000 mL
13. 0.5 L
14. 1500 mL
15. 45 mm
16. 0.6 L
17. 2.5 kg
18. 180 s
19. 120 min
20. 4 min

Medium

1. 1800 mm
2. 40 cm² (Year 8)
3. 2.5 min
4. 4.09 m (Year 8)
5. 32 000 MB
6. 0.25 g
7. 2 500 000 L
8. 0.008 mm
9. 9 kg (Year 8)

Worded problems

1. 64 GB = 64 × 1000 = 64 000 MB; 64 000 ÷ 800 = 80 videos
2. 6 km = 6 × 1000 = 6000 m; 6000 ÷ 400 = 15 laps
3. 135 min = 120 min + 15 min = 2 h 15 min; 7:30 pm + 2 h 15 min = 9:45 pm

10.2 Precision and absolute error

(booklet p. 6)

WE DO examples (teacher)

1. Ex 1a: p = 200 ÷ 4 = 50 g
2. Ex 1b: p = 1 ÷ 10 = 0.1 cm (1 mm); abs. error = 0.1 ÷ 2 = 0.05 cm (0.5 mm)
3. Ex 1c: p = 0.1 kg (last digit); abs. error = 0.1 ÷ 2 = 0.05 kg
4. Ex 1d: Abs. error = 1 ÷ 2 = 0.5 m; limits = 3 − 0.5 to 3 + 0.5 = 2.5 m to 3.5 m
5. Ex 2a: p = 200 ÷ 4 = 50 mL; abs. error = 50 ÷ 2 = 25 mL; limits = 250 − 25 to 250 + 25 = 225 mL to 275 mL
6. Ex 2b: Abs. error = 0.01 ÷ 2 = 0.005 m; limits = 2.45 − 0.005 to 2.45 + 0.005 = 2.445 m to 2.455 m
7. Ex 2c: p = 100 ÷ 5 = 20 g; abs. error = 20 ÷ 2 = 10 g; limits = 340 − 10 to 340 + 10 = 330 g to 350 g
8. Ex 2d: Abs. error = 0.1 ÷ 2 = 0.05 kg; limits = 3.4 − 0.05 to 3.4 + 0.05 = 3.35 kg to 3.45 kg. 3.46 kg is above 3.45 kg, so no.

Easy

1. 0.5 cm
2. 2 kg
3. 5 °C
4. 1 cm
5. 5 mL
6. 0.1 kg
7. 0.1 cm
8. 12.5 mL
9. 1 °C
10. 0.005 m
11. 0.05 kg
12. 0.5 s
13. 5.5 m to 6.5 m
14. 1.45 L to 1.55 L
15. 2.245 kg to 2.255 kg
16. 1.15 kg to 1.25 kg
17. 425 mL to 475 mL
18. 12.5 °C to 17.5 °C

Medium

1. 0.0005 L
2. 150 mm (10.1)
3. 12.445 s to 12.455 s
4. 36.5 °C to 37.5 °C
5. 24 m² (Year 8)
6. 62.5 kg to 67.5 kg
7. 90 min (10.1)
8. (4.15 + 4.25) ÷ 2 = 4.2 m; abs. error = 4.25 − 4.2 = 0.05 m
9. No: the limits are 1.245 m to 1.255 m, and 1.256 m is more than 1.255 m

Worded problems

1. Abs. error = 0.2 ÷ 2 = 0.1 °C; limits = 37.4 − 0.1 to 37.4 + 0.1 = 37.3 °C to 37.5 °C. The true value could be as high as 37.5 °C, so yes: the patient could have a fever.
2. Abs. error = 0.1 ÷ 2 = 0.05 m; each plank 1.45 m to 1.55 m; smallest 1.45 + 1.45 = 2.9 m; largest 1.55 + 1.55 = 3.1 m
3. Sam: abs. error = 0.1 ÷ 2 = 0.05 s, limits 13.35 s to 13.45 s. 13.36 s is inside these limits, so Sam may have been faster: we cannot be sure.

10.3 Percentage error (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: p = 0.1 kg, so E = 0.1 ÷ 2 = 0.05 kg
2. Ex 1b: % error = 0.5 ÷ 20 × 100 = 2.5%
3. Ex 1c: E = 0.1 ÷ 2 = 0.05 L; % error = 0.05 ÷ 2.5 × 100 = 2%
4. Ex 1d: p = 50 mL, so E = 25 mL; % error = 25 ÷ 300 × 100 = 8.3%
5. Ex 2a: E = 1 ÷ 2 = 0.5 m; % error = 0.5 ÷ 24 × 100 = 2.08%
6. Ex 2b: p = 25 mL, so E = 12.5 mL; % error = 12.5 ÷ 175 × 100 = 7.14%
7. Ex 2c: E = 0.01 ÷ 2 = 0.005 kg; % error = 0.005 ÷ 3.25 × 100 = 0.15%
8. Ex 2d: c: it has the smallest percentage error (0.15%)

Easy

1. 0.5 m
2. 0.05 kg
3. 0.005 L
4. 2%
5. 12.5%
6. 10%
7. 5%
8. 2%
9. 0.4%
10. 7.1%
11. 1.4%
12. 8.3%
13. 6.3%
14. 3.6%
15. 5.6%
16. 12.5%
17. 1.9%
18. 2.8%

Medium

1. 1.79%
2. 4 500 g (10.1)
3. 1.35%
4. 0.67%
5. 55 mL to 65 mL (10.2)
6. 2.17%
7. B (A 8.33%, B 4.17%)
8. 0.05 s (10.2)
9. A (A 0.67%, B 2.00%)

Worded problems

1. pencil: 0.5 ÷ 15 × 100 = 3.33%; height: 0.5 ÷ 168 × 100 = 0.30%; her height is more accurate
2. E = 0.1 ÷ 2 = 0.05 s; 0.05 ÷ 12.4 × 100 = 0.40%
3. He used the precision (50 mL), not E = 25 mL. 25 ÷ 750 × 100 = 3.33%

10.4 Perimeter and area of composite shapes (booklet p. 12)

WE DO examples (teacher)

1. Ex 1a: missing sides 10 − 4 = 6 m and 7 − 3 = 4 m; P = 10 + 3 + 6 + 4 + 4 + 7 = 34 m
2. Ex 1b: A = ½ × 8 × 4 = 16 cm²
3. Ex 1c: top part 7 − 3 = 4 cm high; A = 8 × 3 + 3 × 4 = 24 + 12 = 36 cm²
4. Ex 1d: A = 13 × 7 − 11 × 5 = 91 − 55 = 36 m²
5. Ex 2a: the steps add to 12 m across and 9 m up, so P = 12 + 9 + 12 + 9 = 42 m
6. Ex 2b: A = 8 × 5 + ½ × 8 × 3 = 40 + 12 = 52 cm²
7. Ex 2c: triangle base 14 − 8 = 6 m; A = 8 × 5 + ½ × 6 × 5 = 40 + 15 = 55 m²
8. Ex 2d: cut-off corner 12 − 8 = 4 m by 7 − 4 = 3 m; A = 12 × 7 − ½ × 4 × 3 = 84 − 6 = 78 m²

Easy

1. 6 cm
2. 4 m
3. 8 m
4. 30 m
5. 34 cm
6. 36 m
7. 45 cm²
8. 30 m²
9. 20 cm²
10. 30 m²
11. 26 cm²
12. 36 cm²
13. 52 cm²
14. 66 m²
15. 68 m²
16. 34 cm²
17. 40 m²
18. 46 cm²

Medium

1. 28 m
2. 204 cm (10.1)
3. 26 cm²
4. 45 m²
5. 6.35 m to 6.45 m (10.2)
6. 36 cm²
7. 42 m²
8. 32 m
9. 1.25% (10.3)

Worded problems

1. A = 6 × 4 + 3 × 2 = 24 + 6 = 30 m²; 30 × $45 = $1 350
2. missing sides 8 − 2 = 6 m and 6 − 2 = 4 m; P = 8 + 2 + 6 + 4 + 2 + 6 = 28 m; 28 × $15 = $420
3. A = 4 × 2.5 + ½ × 4 × 1.5 − 0.9 × 2 = 10 + 3 − 1.8 = 11.2 m²

10.5 Areas of quadrilaterals (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: $A = bh = 9 \times 5 = 45\ cm^2$
2. Ex 1b: $A = \frac{1}{2}(a + b)h = \frac{1}{2} \times (8 + 14) \times 6 = 66\ cm^2$
3. Ex 1c: $A = \frac{1}{2}xy = \frac{1}{2} \times 10 \times 6 = 30\ m^2$
4. Ex 1d: $A = \frac{1}{2}xy = \frac{1}{2} \times 7 \times 12 = 42\ cm^2$
5. Ex 2a: the 7 cm side is not needed: $A = bh = 10 \times 6 = 60\ cm^2$
6. Ex 2b: the 4 m side is not needed: $A = \frac{1}{2}(a + b)h = \frac{1}{2} \times (4.5 + 7.5) \times 3.2 = 19.2\ m^2$
7. Ex 2c: $A = bh$, so $91 = 13 \times h$, $h = 91 \div 13 = 7$ cm
8. Ex 2d: $A = l \times w + \frac{1}{2}(a + b)h = 10 \times 3 + \frac{1}{2} \times (6 + 10) \times 4 = 30 + 32 = 62\ m^2$

Easy

1. $40\ cm^2$
2. $60\ m^2$
3. $42\ cm^2$
4. $17.5\ m^2$
5. $32\ cm^2$
6. $42\ m^2$
7. $30\ m^2$
8. $50\ cm^2$
9. $66\ cm^2$
10. $35\ m^2$
11. $36\ m^2$
12. $24\ cm^2$
13. $54\ m^2$
14. $25\ cm^2$
15. $30\ cm^2$
16. $44\ m^2$
17. $18\ m^2$
18. $63\ cm^2$
19. $39\ m^2$
20. $26\ cm^2$

Medium

1. $36\ m^2$
2. $52\ m^2$ (10.4)
3. $40\ cm^2$
4. $96\ cm^2$
5. 2% (10.3)
6. 6 cm
7. 4 cm
8. 8.45 m to 8.55 m (10.2)
9. $36\ cm^2$

Worded problems

1. $A = 8.5 \times 6 = 51\ m^2$; $51 \times \$42 = \$2\ 142$
2. $A = \frac{1}{2}xy = \frac{1}{2} \times 80 \times 50 = 2\ 000\ cm^2$
3. $750 = \frac{1}{2} \times (30 + 45) \times h = 37.5 \times h$, so $h = 750 \div 37.5 = 20$ m

10.6 Circumference and arc length

(booklet p. 18)

WE DO examples (teacher)

1. Ex 1a: $C = 2 \times \pi \times 5 = 31.4$ cm
2. Ex 1b: $C = \pi \times 12 = 37.7$ m
3. Ex 1c: $P = \frac{1}{2} \times \pi \times 10 + 10 = 25.7$ cm
4. Ex 1d: $P = \frac{1}{4} \times 2 \times \pi \times 6 + 6 + 6 = 21.4$ m
5. Ex 2a: $l = 60/360 \times 2 \times \pi \times 9 = 9.4$ cm
6. Ex 2b: $P = 9.42\ldots + 9 + 9 = 27.4$ cm
7. Ex 2c: $P = 10 + 6 + 10 + \frac{1}{2} \times \pi \times 6 = 35.4$ m (not the dashed side)
8. Ex 2d: $50 = 2 \times \pi \times r$, so $r = 50 \div (2 \times \pi) = 8.0$ cm

Easy

1. 8 cm
2. 5 m
3. 14 m
4. 4.5 cm
5. 18.8 cm
6. 31.4 m
7. 7.9 m
8. 44.0 cm
9. 12.6 cm
10. 50.3 m
11. 9.4 m
12. 25.1 cm
13. 28.3 m
14. 75.4 mm
15. 20.6 cm
16. 30.8 m
17. 25.7 m
18. 14.3 cm
19. 35.7 m
20. 21.4 cm

Medium

1. 6.3 m
2. $15\ m^2$ (10.5)
3. 12.6 cm
4. 24.6 cm
5. 28.6 cm
6. 25.9 cm
7. 4.45 m to 4.55 m (10.2)
8. 3.0 m
9. 2.5% (10.3)

Worded problems

1. $C = \pi \times 3.2 = 10.05$ m; $10.05 \times \$12 = \$121$
2. $C = 2 \times \pi \times 33 = 207.3$ cm; $100\ 000 \div 207.3 = 482.4$, so 482 full turns
3. $84 + 84 + \pi \times 64 = 369.1$, so 369 m

10.7 Areas of circles and sectors

(booklet p. 21)

WE DO examples (teacher)

1. Ex 1a: $A = \pi \times 5^2 = 78.5\ cm^2$
2. Ex 1b: $r = 12 \div 2 = 6\ m$; $A = \pi \times 6^2 = 113.1\ m^2$
3. Ex 1c: $r = 10 \div 2 = 5\ cm$; $A = ½ \times \pi \times 5^2 = 39.3\ cm^2$
4. Ex 1d: $A = ¼ \times \pi \times 6^2 = 28.3\ m^2$
5. Ex 2a: $A = 120 \div 360 \times \pi \times 9^2 = 84.8\ cm^2$
6. Ex 2b: $A = 10 \times 6 + ½ \times \pi \times 3^2 = 60 + 14.1 = 74.1\ m^2$
7. Ex 2c: $r = 16 \div 2 = 8\ m$; $A = \pi \times 10^2 - \pi \times 8^2 = 314.2 - 201.1 = 113.1\ m^2$
8. Ex 2d: $50 = \pi \times r^2$, so $r^2 = 50 \div \pi = 15.9$ and $r = \sqrt{15.9} = 4.0\ cm$

Easy

1. $28.3\ cm^2$
2. $153.9\ m^2$
3. $78.5\ m^2$
4. $254.5\ cm^2$
5. $50.3\ cm^2$
6. $113.1\ m^2$
7. $3.1\ m^2$
8. $314.2\ cm^2$
9. $380.1\ mm^2$
10. $201.1\ cm^2$
11. $7.1\ m^2$
12. $25.1\ cm^2$
13. $56.5\ m^2$
14. $77.0\ m^2$
15. $12.6\ cm^2$
16. $63.6\ m^2$
17. $19.6\ cm^2$
18. $42.4\ cm^2$
19. $9.4\ m^2$
20. $6.3\ cm^2$

Medium

1. $62.8\ cm^2$
2. $28\ m^2$ (10.5)
3. $47.1\ m^2$
4. $44.6\ cm^2$
5. 30.6 m (10.6)
6. $13.7\ cm^2$
7. $25.4\ cm^2$
8. 41.1 cm (10.6)
9. 3.0 m

Worded problems

1. $A = 120 \div 360 \times \pi \times 6^2 = 37.7\ m^2$
2. $R = 3 + 1 = 4\ m$; $A = \pi \times 4^2 - \pi \times 3^2 = 50.3 - 28.3 = 22.0\ m^2$
3. $r = 30 \div 2 = 15\ cm$; $A = \pi \times 15^2 = 706.9\ cm^2$; $706.9 \div 8 = 88.4\ cm^2$

10.8 Surface area of a prism (booklet p. 24)

WE DO examples (teacher)

1. Ex 1a: the top is l by w: $A = l \times w = 7 \times 3 = 21\ cm^2$
2. Ex 1b: a cube has 6 equal squares: $SA = 6s^2 = 6 \times 5^2 = 6 \times 25 = 150\ m^2$
3. Ex 1c: $SA = 2(lw + lh + wh) = 2 \times (8 \times 5 + 8 \times 4 + 5 \times 4) = 2 \times 92 = 184\ cm^2$
4. Ex 1d: 2 triangles + 3 rectangles: $SA = 8 \times 6 + (8 + 6 + 10) \times 12 = 48 + 288 = 336\ cm^2$
5. Ex 2a: no top: $SA = lw + 2lh + 2wh = 8 \times 5 + 2 \times 8 \times 4 + 2 \times 5 \times 4 = 40 + 64 + 40 = 144\ cm^2$
6. Ex 2b: 2 triangles + 3 rectangles: $SA = 2 \times ½ \times 6 \times 4 + (6 + 5 + 5) \times 9 = 24 + 144 = 168\ cm^2$
7. Ex 2c: inside edges $6 - 2 = 4$ and $5 - 2 = 3$; end: area $6 \times 2 + 2 \times 3 = 18$, perimeter $6 + 2 + 4 + 3 + 2 + 5 = 22$; $SA = 2 \times 18 + 22 \times 8 = 36 + 176 = 212\ cm^2$
8. Ex 2d: $96 = 6s^2$, so $s^2 = 96 \div 6 = 16$ and $s = \sqrt{16} = 4\ cm$

Easy

1. $24\ cm^2$
2. $13.5\ m^2$
3. $25\ cm^2$
4. $35\ m^2$
5. $24\ cm^2$
6. $40\ m^2$
7. $24\ cm^2$
8. $54\ cm^2$
9. $600\ m^2$
10. $37.5\ cm^2$
11. $108\ cm^2$
12. $160\ m^2$
13. $142\ cm^2$
14. $256\ cm^2$
15. $216\ m^2$
16. $46\ m^2$
17. $132\ cm^2$
18. $300\ m^2$
19. $520\ cm^2$

Medium

1. $171\ m^2$
2. $32\ cm^2$ (10.5)
3. $84\ cm^2$
4. $288\ m^2$
5. $113.1\ cm^2$ (10.7)
6. $216\ m^2$
7. 3 m
8. $28\ m^2$ (10.4)
9. $136\ m^2$

Worded problems

1. $SA = 2 \times (30 \times 20 + 30 \times 10 + 20 \times 10) = 2 \times 1\,100 = 2\,200\ cm^2$
2. walls $2 \times (5 \times 2.5 + 4 \times 2.5) = 45\ m^2$, ceiling $5 \times 4 = 20\ m^2$; $45 + 20 = 65\ m^2$; $65 \div 10 = 6.5\ L$
3. $SA = 2 \times ½ \times 3 \times 2 + (3 + 2.5 + 2.5) \times 4 = 6 + 32 = 38\ m^2$

10.9 Surface area of a cylinder (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: $A = \pi \times 3^2 = 28.3\ cm^2$
2. Ex 1b: the rectangle is $2\pi r$ long: $A = 2 \times \pi \times 3 \times 8 = 150.8\ cm^2$
3. Ex 1c: $SA = 2 \times \pi \times 3^2 + 2 \times \pi \times 3 \times 8 = 56.5 + 150.8 = 207.3\ cm^2$
4. Ex 1d: $r = 10 \div 2 = 5$ m; $SA = 2 \times \pi \times 5^2 + 2 \times \pi \times 5 \times 4 = 282.7\ m^2$
5. Ex 2a: one end only: $SA = \pi \times 4^2 + 2 \times \pi \times 4 \times 10 = 50.3 + 251.3 = 301.6\ cm^2$
6. Ex 2b: the label is the curved surface: $r = 7 \div 2 = 3.5$ cm; $A = 2 \times \pi \times 3.5 \times 11 = 241.9\ cm^2$
7. Ex 2c: lying down, h is the length: $r = 2 \div 2 = 1$ m; $SA = 2 \times \pi \times 1^2 + 2 \times \pi \times 1 \times 5 = 6.3 + 31.4 = 37.7\ m^2$
8. Ex 2d: two half circles + half the curved surface + the flat base: $SA = \pi \times 3^2 + \pi \times 3 \times 10 + 6 \times 10 = 28.3 + 94.2 + 60.0 = 182.5\ m^2$

Easy

1. $50.3\ cm^2$
2. $153.9\ m^2$
3. $7.1\ m^2$
4. $314.2\ cm^2$
5. $94.2\ cm^2$
6. $175.9\ m^2$
7. $62.8\ cm^2$
8. $88.0\ cm^2$
9. $131.9\ m^2$
10. $70.7\ cm^2$
11. $100.5\ cm^2$
12. $339.3\ m^2$
13. $326.7\ cm^2$
14. $245.0\ cm^2$
15. $226.2\ m^2$
16. $25.1\ m^2$
17. $125.7\ cm^2$
18. $345.6\ cm^2$
19. $113.1\ m^2$

Medium

1. $103.7\ m^2$
2. 25.7 cm (10.6)
3. $329.9\ cm^2$
4. $219.9\ cm^2$
5. $94\ cm^2$ (10.8)
6. $138.2\ m^2$
7. $452.4\ cm^2$
8. $201.1\ m^2$ (10.7)
9. $255.9\ cm^2$

Worded problems

1. $r = 2 \div 2 = 1$ m; $SA = 2 \times \pi \times 1^2 + 2 \times \pi \times 1 \times 1.5 = 15.7\ m^2$; $15.7 \times \$14 = \$219.80$, so \$220
2. $r = 6 \div 2 = 3$ cm; one turn $= 2 \times \pi \times 3 \times 25 = 471.2\ cm^2$; 10 turns $= 10 \times 2 \times \pi \times 3 \times 25 = 4\ 712.4\ cm^2$
3. $r = 30 \div 2 = 15$ cm; $SA = \pi \times 15^2 + 2 \times \pi \times 15 \times 25 = 706.9 + 2\ 356.2 = 3\ 063.1\ cm^2$

10.10 Volumes of prisms and cylinders

(booklet p. 30)

WE DO examples (teacher)

1. Ex 1a: $V = Ah = 28 \times 4 = 112\ cm^3$
2. Ex 1b: $V = lwh = 8 \times 5 \times 4 = 160\ cm^3$
3. Ex 1c: $V = \frac{1}{2}bh \times l = \frac{1}{2} \times 6 \times 4 \times 7 = 84\ m^3$
4. Ex 1d: $V = \pi r^2 h = \pi \times 3^2 \times 8 = 226.2\ cm^3$
5. Ex 2a: $A = 7 \times 2 + 3 \times 2 = 20\ cm^2$; $V = Ah = 20 \times 4 = 80\ cm^3$
6. Ex 2b: $V = lwh = 50 \times 30 \times 40 = 60\ 000\ cm^3 = 60$ L
7. Ex 2c: $V = lwh$: $96 = 8 \times 3 \times h = 24h$, so $h = 96 \div 24 = 4$ cm
8. Ex 2d: $r = 1$ m; $V = \pi r^2 h = \pi \times 1^2 \times 1.5 = 4.712\ldots\ m^3 = 4\ 712.4$ L

Easy

1. $63\ cm^3$
2. $75\ m^3$
3. $90\ cm^3$
4. $60\ cm^3$
5. $48\ m^3$
6. $30\ m^3$
7. $60\ cm^3$
8. $120\ m^3$
9. $90\ cm^3$
10. $42\ cm^3$
11. $120\ m^3$
12. $54\ cm^3$
13. $62.8\ cm^3$
14. $235.6\ m^3$
15. $28.3\ cm^3$
16. $282.7\ cm^3$
17. $314.2\ m^3$
18. $88.0\ cm^3$

Medium

1. $A = \frac{1}{2} \times (10 + 5) \times 4 = 30\ cm^2$; $V = 30 \times 4 = 120\ cm^3$
2. $r = 5$ m; $A = \pi \times 5^2 = 78.5\ m^2$ (10.7)
3. $A = 8 \times 2 + 3 \times 2 = 22\ m^2$; $V = 22 \times 3 = 66\ m^3$
4. $V = 40 \times 25 \times 30 = 30\ 000\ cm^3 = 30$ L
5. $SA = 2 \times (5 \times 4 + 5 \times 3 + 4 \times 3) = 94\ m^2$ (10.8)
6. $108 = 9 \times 4 \times h = 36h$, so $h = 108 \div 36 = 3$ cm
7. $r = 5$ cm; $V = \pi \times 5^2 \times 8 = 628.3\ cm^3 = 628.3$ mL
8. $SA = 2 \times \pi \times 3^2 + 2 \times \pi \times 3 \times 4 = 131.9\ m^2$ (10.9)
9. $V = \frac{1}{2} \times \pi \times 3^2 \times 10 = 141.4\ m^3$

Worded problems

1. $V = \frac{1}{2}bh \times l = \frac{1}{2} \times 2.4 \times 1.5 \times 3 = 5.4\ m^3$
2. $r = 1.5$ m; $V = \pi r^2 h = \pi \times 1.5^2 \times 2 = 14.137\ldots\ m^3 = 14\ 137$ L
3. $375\ 000$ L $= 375\ m^3$; $h = 375 \div (25 \times 10) = 1.5$ m